SUPERSEDING
NAVORD OS 7290D
22 April 1963

MILITARY SPECIFICATION

EXPLOSIVE FITTING MARK 7 MOD 0

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This specification covers the requirements for the procurement of the Explosive Fitting Mark 7 Mod 0 (Empty), Bureau of Naval Weapons LD No. 299474 and Explosive Fitting Mark 7 Mod 0 (Explosive Loaded), Bureau of Naval Weapons LD No. 479589. The Explosive Fitting Mark 7 Mod 0 is electrically fired to release a signal device used in drill mines. It is of one type and one class.

2. APPLICABLE DOCUMENTS

2.1 The following standards, drawings, and publications or such portions thereof as designated herein of the issue in effect on the date of invitation for bids form a part of this specification.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-202	Test Methods for Electronic and Electric Component Parts
MIL-STD-350	Jolt Test for Use in Production of Fuzes

FSC 1350 (Empty)
FSC 1351 (Loaded)

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MIL-STD-352	Forty (40) Foot Drop Test for Use in Production of Fuzes
MIL-STD-353	Transportation Vibration Test for Use in Production of Fuzes
MIL-STD-354	Temperature and Humidity Test for Use in Production of Fuzes

DRAWINGS

BUREAU OF NAVAL WEAPONS

LD 267056	Test Set Mark 161 Mod 0
LD 267058	Test Set Mark 162 Mod 0
LD 267280	Test Set Mark 238 Mod 0
LD 284578	Test Chamber Mark 4 Mod 0
LD 284775	Explosive Actuator Mark 1 Mod 0
LD 299474	Explosive Fitting Mark 7 Mod 0 (Empty)
LD 479589	Explosive Fitting Mark 7 Mod 0 (Explosive Loaded)
LD 479680	Container, Shipping, for Explosive Fitting Mark 7 Type
1476194	Marking Instructions for Shipping and Storage Containers (Inner and Outer)
1664860	Jolt Fixture
1664901	Vibration Fixture

And all drawings, standards and specifications listed thereon. In the following specification, all referenced drawing numbers are listed on Bureau of Naval Weapons LD 479589 unless otherwise designated.

CLASSIFICATION OF DEFECTS

BUREAU OF NAVAL WEAPONS

OCD 1299587	Explosive Fitting Mark 7 Mod 0

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PUBLICATIONS

BUREAU OF NAVAL WEAPONS

OD 6819	Test Set Mark 161 Mod 0
OD 6821	Test Set Mark 152 Mod 0
OD 6854	Test Set Mark 238 Mod 0
OD 10511	Basic Methods for Nonmagnetic Testing with the Test Set Mark 115 Type

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78	Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1946 Edition - Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950.)"

NATIONAL BUREAU OF STANDARD PUBLICATIONS

Handbook H28	Screw Thread Standards for the Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Description. - The Explosive Fitting Mark 7 Mod 0 consists of a substantial stainless steel housing confining and directing the force produced by firing the Explosive Actuator Mark 1 Mod 0 and an electrical cable and plug used in initiating the firing. One end of the steel housing is threaded for attachment of the Explosive Fitting Mark 7 Mod 0 in service.

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3.2 General requirements.

3.2.1 Preproduction samples. - Unless otherwise specified in the contract or order preproduction samples of the Explosive Fitting Mark 7 Mod 0 shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification and the design. Provision of samples shall be in accordance with 4.1.2.

3.2.2 Materials. - Unless otherwise specified in the contract, requisition or order all materials and purchased parts entering into the final assembly of the Explosive Fitting Mark 7 Mod 0 shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BUORD DL 479589 and BUORD LD 299474.

3.2.3 Materials, finishes, coatings, construction and dimensions. - Materials, finishes, coatings, construction and dimensions for the explosive fitting shall meet the requirements of LD 479589 and this specification. Examination shall be in accordance with 4.5.4.1.

3.2.4 Marking. - Marking of assemblies, sub-assemblies and parts shall be in accordance with the applicable drawings listed in BUORD LD Nos. 479589 and 299474.

3.2.5 Interchangeability. - Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in strict agreement with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings will be made without prior revisions and correction of the applicable drawings.

3.2.6 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of the pertinent Naval Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility, in the design and manufacture of such gages as may be required

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in the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Naval Final Inspection Gages.

3.2.7 Threads. - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings and the NOT-GO gaging practice shall be as set forth in the aforementioned Handbook H28.

3.2.8 Conflicting requirements. - Conflicting requirements arising between this specification or any specification, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.2.9 Explosive Fitting Mark 7 Mod 0 (Explosive Loaded).- Requirements, product characteristics and tests for the Explosive Fitting Mark 7 Mod 0 (Empty) shall apply to the Explosive Fitting Mark 7 Mod 0 (Explosive Loaded).

3.3 Definitions. - The following terms will be used throughout this specification on the Explosive Fitting Mark 7 Mod 0 and are here defined for clarity.

3.3.1 Explosive fitting. - Reference to the "explosive fitting" shall mean the Explosive Fitting Mark 7 Mod 0 (Empty) or the Explosive Fitting Mark 7 Mod 0 (Explosive Loaded), depending upon which is subject to procurement.

3.3.2 Empty explosive fitting. - Reference to the "empty explosive fitting" shall mean the Explosive Fitting Mark 7 Mod 0 (Empty) as covered by BUORD LD 299474.

3.3.3 Loaded explosive fitting. - Reference to the "loaded explosive fitting" shall mean Explosive Fitting Mark 7 Mod 0 (Explosive Loaded) as covered by BUORD LD 479589.

3.4 Performance requirements and product characteristics. - The explosive fitting shall meet the following performance requirements and product characteristics:

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3.4.1 Cleanliness. - All components of the explosive fitting shall be free of moisture, corrosion, dirt, dust, metal chips, filings or other such foreign matter at the time of assembly and packaging. The internal explosive cavity shall be carefully blown out immediately prior to potting.

3.4.2 Soldering. - All connections requiring soldering shall use solder and flux as specified on the drawings. The connections shall be mechanically secure before soldering. All loose and excess solder shall be removed immediately after each soldering operation. All flux shall be removed.

3.4.3 Cable securement. - The electrical cable of the loaded explosive fitting shall be securely attached at both ends to withstand a direct longitudinal pull as indicated in 4.5.4.2 of 10 lbs. minimum for a period of 10 seconds minimum without showing any evidence of loosening at either end. The explosive fitting shall also be capable of meeting the insulation resistance and continuity requirements after being subjected to the pull above.

3.4.4 Insulation resistance. - When tested in accordance with 4.5.4.3, the insulation resistance of the cable portion of the empty explosive fitting and the loaded explosive fitting shall be 50 megohms minimum.

3.4.5 Continuity resistance. - When tested in accordance with 4.5.4.4 the continuity resistance of the empty explosive fitting shall not exceed 0.2 ohms between each pin of the electrical connector and the corresponding terminal at the other end of the electrical cable.

3.4.6 Static magnetic effect. - When tested in accordance with paragraph 4.5.4.5, the static magnetic effect of the empty explosive fitting shall not be great enough to cause a total change in flux density of the background magnetic field of more than 0.10 milligauss (peak to trough). When the loaded explosive fitting is procured the static magnetic effect test shall be performed prior to loading.

3.4.7 Loaded explosive fitting. - In assembly of the loaded explosive fitting, the requirements and tests of BUORD LD 284775, Explosive Actuator Mark 1 Mod 0 shall apply. The Actuator Mark 1 Mod 0 shall have been manufactured and tested within six (6) months prior to assembly or retested in accordance with BUORD Dwg. 1170774, a part of BUORD LD 284775, prior to use.

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3.4.8 Final assembly continuity resistance. - When the loaded explosive fitting is submitted to the final continuity resistance test of 4.5.4.6, utilizing Test Set Mark 238 in accordance with OD 6854 the resistance shall be not less than 3.0 ohms or greater than 7.0 ohms. During this test the loaded explosive fitting shall be mounted in Test Chamber Mark 4 Mod 0 and the current shall not exceed 5 milliamperes. The resistance shall be measured to an accuracy of ± 0.1 ohm.

3.4.9 High pressure requirement. - The loaded explosive fitting shall meet all requirements of this specification after a hydrostatic pressure of 550 pounds per square inch has been applied to the sea side of the fitting as indicated in 4.5.4.7.

3.4.10 Functioning. - The loaded explosive fitting shall function when subjected to the discharge of a 1.0 ± .02 microfarad condenser charged to 38.5 volts. The discharge shall be applied by means of Test Set Mark 152 Mod 0 which includes a 4 ohm resistor in series with the explosive fitting. Acceptance and testing shall be in accordance with 4.5.4.8.

3.5 Environmental requirements.

3.5.1 Vibration

3.5.1.1 Vibration, transportation, storage and handling. - The loaded explosive fitting shall be subjected to the vibration test of 4.6.1.1 and thereafter meet the requirements of 3.2.3, 3.4.4, 3.4.8, 3.4.9, and 3.4.10.

3.5.1.2 Vibration, ready-non-operating. - The loaded explosive fitting shall be subjected to the vibration test of 4.6.1.2 and thereafter meet the requirements of 3.2.3, 3.4.4, 3.4.5, 3.4.8, 3.4.9, and 3.4.10.

3.5.2 Shock.

3.5.2.1 Shock, free fall drop. - The loaded explosive fitting shall be subjected to the shock test of 4.6.2.1 and therafter meet the requirements of 3.2.3, 3.4.4, 3.4.8, 3.4.9, and 3.4.10.

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3.5.2.2 Shock, ready-non-operating. - The loaded explosive fitting shall be subjected to the shock test of 4.6.2.2 and thereafter meet the requirements of 3.2.3, 3.4.4, 3.4.5, 3.4.8, 3.4.9 and 3.4.10.

3.5.3 Temperature and humidity. - The loaded explosive fitting shall be subjected to the temperature test of 4.6.3 and thereafter meet the requirements of 3.2.3, 3.4.4, 3.4.5, 3.4.8, 3.4.9 and 3.4.10.

3.5.4 Safety requirements.

3.5.4.1 Jolt requirement. - The loaded explosive fitting shall be subjected to the jolt test of 4.6.4.1 and thereafter shall be safe to handle and dispose of and shall not have fired.

3.5.4.2 Forty foot drop requirement. - The loaded explosive fitting shall be subjected to the 40 foot drop test specified in 4.6.4.2 and thereafter shall be safe to handle and dispose of and shall not have fired.

3.6 Clean-up. - Prior to and after final assembly all parts, components and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.7 Workmanship. - The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, machine screw assemblage, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Government inspection. - The Government representative will examine the contractor's records and processes and will witness or perform examinations deemed necessary to assure that components and assemblies of the Explosive Fitting Mark 7 Mod 0 are in accordance with all applicable requirements.

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4.1.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products shall be in accordance with the provisions set forth in MIL-STD-105.

4.1.2 Preproduction sample. - Unless otherwise specified for each contract or order, a representative sample of 219 explosive fittings shall be manufactured using the methods and procedures proposed for the production lot. These shall be submitted to the Government inspector for inspection to determine conformance with the requirements of 3.4, using tests of 4.5, and a lot size of 2500 as the basis for inspection. After completion of the acceptance tests of 4.5, the remainder of the preproduction sample together with a report of inspection shall be forwarded to an activity designated in the contract or by the Bureau of Naval Weapons for subjection to the tests of 4.6. The tests of 4.5 and 4.6, which are performed on the preproduction sample are for purposes of determining compliance of the preproduction sample with the requirements of the contract, specifications and drawings. Further production of the item by the contractor prior to approval of the contracting activity or completion of the test on the preproduction sample shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order. The preproduction sample shall be allocated in accordance with Table I.

4.1.3 Lot size. - As applied to Government inspection of units of product the term "lot" shall mean "Inspection lot", i.e., a collection of units of product submitted by a supplier for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" will be as determined by the Government inspector and may differ from the quantity designated in the contract or order in a lot for production, shipment or other purpose.

4.1.4 Periodic production sample. - From each acceptable lot or grand lot of 10,000 maximum, a random sample of 55 explosive fittings will be selected by the Government inspector for submission to a Bureau of Naval Weapons designated activity to determine compliance of the samples with the requirements of the contract, specifications and drawings. The sample shall be considered satisfactory if it meets the requirements of 3.4 and 3.5. The periodic production sample shall be allocated in accordance with Table II.

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4.1.5 Production lot. - For procurement purposes each production lot shall consist of 2000 explosive fittings plus those required for all test purposes. Unless otherwise stated in the contract, order or requisition, all units for test purposes shall be at the expense of the contractor. The production lot should not be confused with the lot for inspection purposes as defined in 4.1.3.

4.2 Inspection.

4.2.1 Contractor inspection. - Unless otherwise specified by the procuring activity, the contractor is responsible for the performance of all tests and examinations to assure that Explosive Fittings Mark 7 Mod 0 are delivered to the Government in full accordance with all applicable requirements. In addition, the contractor is required to maintain records of all tests and examinations performed and to make these available to the Government representative upon request.

4.3 Classification of tests. - The inspection and testing of the explosive fitting shall be classified as follows:

4.3.1 Acceptance tests. - These tests are to be accomplished on explosive fittings being submitted for acceptance under contract. Inspection and test procedures shall be submitted for Government approval prior to commencement of production. Acceptance tests shall be performed by the manufacturer. These tests are defined in 4.5.

4.3.2 Preproduction and periodic production tests. - Preproduction and periodic production tests are those which are accomplished on samples selected as specified in 4.1.2 and 4.1.4 which are representative of the production of the explosive fitting after the award of the contract to determine that the lot and production meet the requirements of this specification. These are detailed in 4.6. Acceptance shall be based on no defects in a sample. Failure of a sample to comply with these requirements will result in the rejection of the lot or the cessation of production as determined by the procuring activity.

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TABLE I

ALLOCATION OF PREPRODUCTION SAMPLES FOR TESTING

Unit	Test	Site of Test	No. of Units	Notes
Empty explosive fitting	4.5.4.2 (Destructive)	Manuf. Plant	15	This test is performed prior to assembly of preproduction sample.
Empty explosive fitting	4.5.4.3 (Non-Destructive)	Manuf. Plant	219	This test is performed prior to assembly of preproduction sample.
Empty explosive fitting	4.5.4.4 (Non-Destructive)	Manuf. Plant	219	This test is performed prior to assembly of preproduction sample.
Empty explosive fitting	4.5.4.5 (Non-Destructive)	Manuf. Plant	219	This test is performed prior to assembly of preproduction sample.
Loaded explosive fitting	4.5.4.6 (Non-Destructive)	Manuf. Plant	219	
Loaded explosive fitting	4.5.4.7 (Non-Destructive)	Manuf. Plant	219	
Loaded explosive fitting	4.5.4.8 (Destructive)	Manuf. Plant	60 - 140	

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TABLE I (Contd)

ALLOCATION OF PREPRODUCTION SAMPLES FOR TESTING

Unit	Test	Site of Test	No. of Units	Notes
Loaded explosive fitting	4.6.1.1 (Destructive)	Designated by contract or procuring agency	10	
Loaded explosive fitting	4.6.1.2 (Destructive)	Designated by contract or procuring agency	10	
Loaded explosive fitting	4.6.2.1 (Destructive)	Designated by contract or procuring agency	24	
Loaded explosive fitting	4.6.2.2 (Destructive)	Designated by contract or procuring agency	10	
Loaded explosive fitting	4.6.3 (Destructive)	Designated by contract or procuring agency	10	
Loaded explosive fitting	4.6.4.1 (Destructive)	Designated by contract or procuring agency	10	
Loaded explosive fitting	4.6.4.2 (Destructive)	Designated by contract or procuring agency	5	

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TABLE II

ALLOCATION OF PERIODIC PRODUCTION SAMPLES FOR TESTING

Unit	Test	Site of Test	No. of Units	Notes
Loaded explosive fitting	4.6.1.2 (Destructive)	Designated by contract or procuring agency	10	
Loaded explosive fitting	4.6.2.2 (Destructive)	Designated by contract or procuring agency	10	
Loaded explosive fitting	4.6.3 (Destructive)	Designated by contract or procuring agency	20	
Loaded explosive fitting	4.6.4.1 (Destructive)	Designated by contract or procuring agency	10	
Loaded explosive fitting	4.6.4.2 (Destructive)	Designated by contract or procuring agency	5	

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4.4 Test equipment. - The following items of test equipment are required to perform the acceptance tests set forth in this specification:

4.4.1 Standard measuring instruments together with

Test Set Mark 238 Mod 0	LD 267058
Test Set Mark 161 Mod 0	LD 267056
Test Set Mark 152 Mod 0	LD 267280
Test Chamber Mark 4 Mod 0	LD 284578
Test Set Mark 115 Mod 4	Government furnished

4.5 Acceptance tests.

4.5.1 The contractor shall conduct acceptance tests as specified in 4.3.1 to assure that the explosive fitting is in compliance with the requirements of 3.2 and 3.4.

4.5.2 Test conditions. - Unless otherwise specified the explosive fitting shall be subjected to acceptance tests under the following conditions:

4.5.2.1 Temperature: Room ambient 65 degrees F to 95 degrees F (18.33 degrees C to 35.0 degrees C).

4.5.2.2 Altitude: Normal ground.

4.5.2.3 Humidity: Room ambient to 95 percent relative, maximum.

4.5.3 Test and inspection equipment and facilities. - The manufacturer shall furnish and maintain all necessary test equipment, except that furnished by the Government for which maintenance is required, facilities and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity, and when definite requirements are not specified they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.5.4 Test procedures.

4.5.4.1 Examination. - The explosive fitting assemblies shall be examined to determine conformance with this specification and LD 479589 and all drawings listed thereon. Examination shall be conducted in accordance with OCD 1299587. Acceptance and rejection shall be in accordance with MIL-STD-105 and the stated AQL.

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4.5.4.2 Cable securement test. - The loaded explosive fitting shall be attached to a test fixture which holds each end fitting in a manner similar to the method by which they are attached in service. The load shall be applied so that the resultant force causes a longitudinal pull of 10 ± 1 pounds in the cable tending to withdraw it from the end fittings. This force shall be applied for a period of 10 seconds minimum. During and after the test examine carefully the cable area adjacent to each end fitting for evidence of slippage of the securing means. Any evidence of slippage shall be cause for regarding the explosive fitting defective. Sampling shall be in accordance with MIL-STD-105, Inspection Level 1, 1.0% AQL. Following this test the loaded explosive fitting shall be subjected to the insulation resistance test and the final assembly continuity resistance test.

4.5.4.3 Insulation resistance test. - The insulation resistance test shall be made on the cable portion of the empty explosive fitting and the loaded explosive fitting. The test on the cable shall be made between leads and between the two leads shunted together and the connector body. The test on the loaded explosive fitting shall be made between the two leads shunted together and the housing connected to ground. The test apparatus to be used for this test shall be Test Set Mark 161 Mod 0 (Insulation, Resistance). Operating instructions for the test set are given in NAVORD OD 6819. The temperature and relative humidity during the test shall be less than 79° F and 80 percent, respectively. To facilitate production, ambient conditions may exceed the prescribed limits for acceptance but rejection of material shall be based on tests conducted as prescribed. An acceptable assembly shall have a resistance of 50 megohms or greater between the lead wires when a potential of 1000 volts a.c. (RMS) is applied for a period of not less than one second.

4.5.4.4 Continuity resistance. - The electrical resistance of the cable portion of the empty explosive fitting shall be tested in accordance with MIL-STD-202, Method 303, between each pin of the electrical connector and the corresponding terminal at the opposite end of the electrical cable. Any empty explosive fitting failing to meet the requirements of 3.4.5 shall be regarded as defective.

4.5.4.5 Static magnetic effect test. - The magnetic testing of the Explosive Fitting Mark 7 Mod 0 shall be conducted prior to explosive loading and with the Test Set Mark 115 Mod 4 in accordance with Test Method 1 of NAVORD OD 10511 under the general test conditions and definitions described therein. For this test the fitting shall be prepared by forming into a flat coil shape with the explosive plug and plug holder arranged in the center of the coil. The diameter of the coil shall not exceed 5 inches. Cord, tape or the like shall be used where necessary to secure the assembly against relative movement

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during testing. Either flat surface of the coil may be selected as the test face. The distance from the center of the magnetometer to the test face of the explosive fitting shall be 1-3/4 ± 1/16 inch. The rate at which the explosive fitting shall be rotated when conducting the test of orientation (C) shall be between 5 and 15 seconds per revolution. When tested as prescribed, an explosive fitting producing a magnetic signal exceeding 0.10 milligauss when evaluated in accordance with section 2.5 of OD 10511 shall be classed as defective.

4.5.4.6 Final assembly continuity resistance test. - The loaded explosive fitting shall be mounted in Test Chamber Mark 4 Mod 0 and shall have a resistance between the limits of 3.0 and 7.0 ohms when tested in accordance to the operating instructions for Test Set Mark 238 Mod 0. Measuring current shall not exceed 5 milliamperes. Resistance which is found beyond these limits shall be cause for rejection.

4.5.4.7 High pressure test. - Mount the loaded explosive fitting within a suitable pressure pot and apply 550 psi hydrostatic pressure to the housing, cable and sea side of the connector. After a minimum of three hours, release the pressure and subject the loaded explosive fitting to the insulation resistance test of 4.5.4.3 and the final assembly continuity test of 4.5.4.6. Any loaded explosive fitting failing to meet any requirement of this specification after completion of this test shall be considered defective.

4.5.4.8 Functioning test. - Samples shall be selected and tested from each lot in accordance with multiple sampling plan, Inspection Level I, AQL of 0.65 percent defective as given in MIL-STD-105. Leads from the explosive fitting shall be attached to the terminals of Test Set Mark 152 Mod 0. The leads between the test set and the firing chamber shall have the shortest practicable length. Adjust the capacitor voltage of the test set to 36.5 volts, and close the firing circuit by means of the switch. (See OD 6821) Explosive fittings which do not explode on receiving the first capacitor discharge shall be considered defectives. Disposition of the lot shall be governed by the applicable sampling table.

4.6 Preproduction and periodic production tests. - The following tests are to be performed at a Bureau of Naval Weapons designated activity and shall not be performed by the contractor unless authorized by the Bureau of Naval Weapons.

4.6.1 Vibration.

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4.6.1.1 Vibration, transportation, storage and handling. - The loaded explosive fitting, on preproduction tests only, packaged and packed in shipping containers as indicated in section 5, shall be mounted on a vibration platform equipped with barriers so as to allow free movement of the container relative to the platform but prevent the devices from falling off the platform. The motion of the platform shall be such that any point on the platform moves in a circular path one inch in diameter with the plane of the circle perpendicular to the platform. The frequency shall be within the range of 240 to 300 cycles per minute. The package of units shall be vibrated for a period of one half-hour on each face of the container. After completing the above vibration test, the loaded explosive fitting shall be capable of meeting the requirements of 3.2.3, 3.4.4, 3.4.8, 3.4.9 and 3.4.10.

4.6.1.2 Transportation vibration test. - The loaded explosive fitting shall be mounted rigidly to the vibration table by means of Vibration Fixture, BUWEPS Dwg. 1664901, and shall be subjected to the transportation vibration test defined in MIL-STD-353. After completion of the vibration test, the loaded explosive fitting shall not have fired or become unsafe to handle or use and shall be capable of meeting the requirements of 3.2.3, 3.4.4, 3.4.8, 3.4.9 and 3.4.10.

4.6.2 Shock.

4.6.2.1 Shock, free fall drop test. - On preproduction samples only, the loaded explosive fitting packaged and packed as designated in 5, shall be subjected to a free fall drop shock from a height of thirty inches to a concrete or hard steel surface. This test shall be applied to each corner of the packaged unit. Upon completion of the test the loaded explosive fitting shall be capable of meeting the requirements of 3.2.3, 3.4.4, 3.4.8, 3.4.9 and 3.4.10.

4.6.2.2 Shock, ready-non-operating. - The loaded explosive fitting shall be subjected to a continuous two-phase shock applied in any direction with respect to the mechanism.

A. The first (impact) phase shall consist of a change of velocity of 5.5 to 7.0 feet per second taking place in from 0.2 to 0.4 millisecond.

B. During the second (drag) phase, the device shall be accelerated for a minimum time of 15 milliseconds in such a manner that during this time the minimum average acceleration is 100 gravity units.

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The peak acceleration shall be between 125 and 150 gravity units. Upon completion of the above test the fitting shall be capable of meeting the requirements of 3.2.3, 3.4.4, 3.4.5, 3.4.8, 3.4.9 and 3.4.10.

4.6.3 Temperature and humidity test. - Twenty explosive fittings shall be subjected to the temperature cycling test specified in MIL-STD-354. After completing the said test and at ambient temperature, all explosive fittings shall meet the requirements of 3.2.3, 3.4.4, 3.4.5, 3.4.8, 3.4.9 and 3.4.10.

4.6.4 Safety tests.

4.6.4.1 Jolt test. - The loaded explosive fittings shall be mounted in Jolt Test Fixture, BUORD Dwg. 1664860, and subjected to the jolt test defined in MIL-STD-350. After completion of the test any fitting that has fired or that has become unsafe to handle shall be regarded as defective.

4.6.4.2 Forty foot drop test. - The loaded explosive fitting shall be subjected to the drop test indicated in MIL-STD-352. After completion of the test any fitting that has fired or that has become unsafe to handle shall be regarded as defective.

4.7 Acceptance criteria.

4.7.1 Preproduction. - Failure of the explosive fitting to meet all the requirements and tests of this specification shall be cause for regarding the explosive fitting as defective and shall result in the rejection of the preproduction sample.

4.7.2 Production acceptance. - Failure of the explosive fitting to meet all the requirements and tests of this specification shall be cause for regarding the explosive fitting as defective and shall result in the rejection of the lot, based on the AQL as noted in OCD 1299587.

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4.7.3 Periodic. - Failure of the explosive fitting to meet all the requirements and tests of this specification shall be cause for regarding the explosive fitting as defective and shall result in the rejection of the lot from which the sample was taken.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.

5.1.1 Level A.

5.1.1.1 Twenty-four explosive fittings shall be packaged and packed together in a metal container in accordance with BUORD LD No. 479680.

5.2 Marking.

5.2.1 In addition to any special marking required by contract or order, all marking shall be in accordance with Military Standard MIL-STD-129, Bureau of Naval Weapons Drawing 1476194, and Code of Federal Regulations 49 CFR 71-78.

6. NOTES

6.1 Intended use. - The explosive fitting is intended for use in the Drill Mine Mark 56 Type.

6.2 Ordering data. - Procurement documents should specify Level A packaging and packing requirements for all shipments.

6.3 General safety precautions. - The loading, assembly and handling of explosives, ammunition, ammunition components, fuzes, detonators, primers, tracers, boosters, pyrotechnics, etc., and subassemblies thereof, involve hazardous operations and therefore require suitable explosives safety precautions. Use of this specification shall not be construed as to relieve the contractor or manufacturer of responsibility for the safety of his operations. Listed below are the minimum provisions which a contractor or manufacturer (loader) of the above items should observe in order to fulfill his responsibility for safety. At Bureau of Naval Weapons, Navy Department, and other Government plants, these provisions are mandatory. Such other warnings and precautions, pertinent to the operational effectiveness or safety during use or loading of the specified item, are included in the detail provisions of the specification.

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6.3.1 All loading operations should be conducted in a neat and orderly manner.

6.3.2 Safe equipment and methods should be utilized for transporting and handling explosives and loaded parts. Where required, remote control barricaded handling equipment shall be used for explosive operations, such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

6.3.3 Personnel handling detonators, primers, delay elements, lead-ins, boosters, and related parts which effect functioning, should in so far as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.3.4 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.3.5 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance with the American Table of Distances (A.T.D.) or other applicable safety standards; and, while in process, in safety lockers and chests if in loading rooms, or in adequate ready or service magazines located in accordance with intraplant distances when outside of loading rooms. For Navy explosives loading plants, the provisions of the Armed Services Explosives Safety Board covering quantity-distance relations for explosives, shall apply.

6.3.6 Proper care should be exercised at all times to protect personnel from accidents, fires or explosions, and to limit damage to equipment and loading areas. In this connection, the following precautionary measures should be observed:

6.3.6.1 Employ properly proportioned and properly located protective barricades, screens or shields at all required points.

6.3.6.2 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each state of operation.

6.3.6.3 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges.

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6.3.6.4 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges should be employed. Fabrics such as silk and nylon, which promote static generation should be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items which are unusually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low-energy type electric primers, detonators and squibs. The latter types of items should have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare non-insulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding of these items from induced electric currents generated by sources such as lightning, static, radiations from communications apparatus, radar, or high frequency heating apparatus, etc. Where necessary for safety, humidity of work rooms should be increased, as required to lessen electrostatic effects without excessive moisture absorption.

6.3.6.5 Protect all explosive operations from effects of electric current originating from equipment such as soldering irons, heaters, switches, wiring, motors, lights, test instruments, etc., by suitable insulation, grounding, separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or due to completing an electric circuit through an electric primer, detonator, or squib. Circuits may be inadvertently completed, for example, from a defective electric soldering iron through a grounded contact. All electric type primers, detonators or squibs provided with wire leads should have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Where practicable, removable short circuiting clips, or other devices should be employed during manufacturing operations involving electric primers, detonators or squibs.

6.3.6.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning and associated industrial hazards.

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6.3.6.7 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.3.6.8 Enforce good housekeeping and maintain effective policing inspection and supervisory methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from floors, walls, ceilings, ledges, tables, benches, piping, and equipment or the items loaded; also, clean up any spilled material immediately.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 49-R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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